SEQUENTIAL CIRCUITS & FLIP-FLOPS

Complete High-Yield Engineering Reference Handbook & Study Guide

SOURCE: R.P. JAIN (CH 7 & CH 8) | TARGET: ADE / DEC / VLSI

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1. Foundations of Sequential Logic & Latches

Unlike **combinational circuits** whose output depends purely on present input levels, a **sequential digital circuit** utilizes memory elements and feedback paths. Its output is a function of both the present external inputs and the **present state** (stored past history).

ATTRIBUTE	COMBINATIONAL CIRCUIT	SEQUENTIAL CIRCUIT
Output Dependency	Present inputs only: \(Y = f(X)\)	Present inputs + Past states: \(Y = f(X, Q_n)\)
Memory Elements	No memory required (direct gate logic)	Latches or Flip-Flops present
Feedback Paths	Strictly feedforward (no feedback)	Output fed back to combinational inputs
Clock Synchronization	Clockless	Synchronous (clocked) or Asynchronous
Operating Speed	Faster (governed only by gate delays)	Governed by clock cycle & setup/hold constraints

1.1 The Basic 1-Bit Memory Cell & Latches

The fundamental 1-bit memory cell consists of two cross-coupled inverters. To enter data externally, 2-input **NAND** or **NOR** gates are used, forming the **S-R Latch**.

NAND S-R Latch (Active-Low Inputs)

\(□AR{S}\)	\(□AR{R}\)	\(Q_{N+1}\)	STATE / ACTION
0	0	**1 / 1 (?)**	**Forbidden / Invalid**
0	1	**1**	Set State (\(Q=1\))
1	0	0	Reset State (\(Q=0\))
1	1	\(Q_n\)	No Change (Memory Hold)

NOR S-R Latch (Active-High Inputs)

\(S\)	\(R\)	\(Q_{N+1}\)	STATE / ACTION
0	0	\(Q_n\)	No Change (Memory Hold)
0	1	0	Reset State (\(Q=0\))
1	0	**1**	Set State (\(Q=1\))
1	1	**0 / 0 (?)**	**Forbidden / Invalid**

> ⚠️ **Hardware Rule: Active-Low Preset (\(□ar{Pr}\)) and Clear (\(□ar{Cr}\))**
>
> Most practical TTL ICs (e.g. **7474, 7476**) feature asynchronous \(□ar{Pr}\) and \(□ar{Cr}\) pins. Because they operate directly on the internal cross-coupled latch, they **override the clock and data inputs**. To operate normally in synchronous mode, both \(□ar{Pr}\) and \(□ar{Cr}\) must be tied to logic **1 (\(V_{CC}\))**.

2. The Core 4 Flip-Flops: Characteristics & Truth Tables

2.1 Clocked S-R Flip-Flop

Gating the S-R inputs with a clock signal ensures transitions only occur during active clock intervals.

\(CK\)	\(S\)	\(R\)	\(Q_N\)	\(Q_{N+1}\)	ACTION / MODE
0	\(imes\)	\(imes\)	\(Q\)	\(Q\)	No Clock (Latched)
1	0	0	\(Q\)	\(Q\)	No Change (Hold)
1	0	1	\(imes\)	0	Reset
1	1	0	\(imes\)	1	Set
1	1	1	\(imes\)	**?**	**Prohibited State (\(S \cdot R = 0\) violation)**

> 📐 **Characteristic Equation: S-R Flip-Flop**
>
> **\(Q_{n+1} = S + \bar{R}Q_n\)** (Subject to constraint: \(S \cdot R = 0\))

2.2 J-K Flip-Flop & The Race-Around Condition

The J-K flip-flop solves the forbidden state of the S-R flip-flop by feeding back the outputs: \(S = J\bar{Q}\) and \(R = KQ\). When \(J = K = 1\), the output **toggles** on every clock pulse.

\(J\)	\(K\)	\(Q_N\)	\(Q_{N+1}\)	OPERATION
0	0	\(Q\)	\(Q\)	Hold / No Change
0	1	\(imes\)	0	Reset
1	0	\(imes\)	1	Set
1	1	\(Q\)	**\(\bar{Q}\)**	**Toggle Mode**

> 📐 **Characteristic Equation: J-K Flip-Flop**
>
> **\(Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n\)**

> ⚡ **The Race-Around Condition (Critical Exam & Viva Topic)**
>
> **Cause:** In a level-triggered J-K flip-flop with \(J=K=1\), if the clock pulse duration \(t_p\) is longer than the flip-flop's internal propagation delay \(\Delta t\) (\(t_p > \Delta t\)), the output toggles repeatedly during the single high pulse. At the end of the clock pulse, the final state of \(Q\) is unpredictable and indeterminate.
> **Condition to Avoid:** \(t_p < \Delta t < T\) (Practically difficult with high-speed ICs).
> **Solution:** 1. **Master-Slave J-K Flip-Flop (e.g. IC 7476)**: Isolates input reading on master during clock HIGH, and transfers to slave on falling edge. 2. **Edge-Triggered Flip-Flops (e.g. IC 7474, 74LS112)**: Responds strictly at the rising or falling edge of the clock (data lockout).

2.3 D-Type Flip-Flop (Data / Delay)

Formed by setting \(K = \bar{J}\) or \(R = \bar{S}\). Transfers data directly to output on clock trigger.

\(D\)	\(Q_{N+1}\)	OPERATION
0	0	Reset
1	1	Set

\(Q_{n+1} = D\) (Used in registers, pipeline buffers, memory cells)

2.4 T-Type Flip-Flop (Toggle)

Formed by tying \(J = K = T\). Toggles output when \(T = 1\), holds state when \(T = 0\).

\(T\)	\(Q_{N+1}\)	OPERATION
0	\(Q_n\)	Hold
1	\(\bar{Q}_n\)	Toggle

\(Q_{n+1} = T \oplus Q_n\) (Used in binary ripple & sync counters)

3. The Master Excitation Table & Flip-Flop Conversions

An **Excitation Table** specifies the required inputs to cause a desired state transition from **Present State (\(Q_n\))** to **Next State (\(Q_{n+1}\))**. It is the core mathematical tool for designing counters, sequence generators, and FSMs.

STATE TRANSITION		S-R FLIP-FLOP		J-K FLIP-FLOP		T FLIP-FLOP	D FLIP-FLOP
PRESENT (\(Q_N\))	NEXT (\(Q_{N+1}\))	\(S\)	\(R\)	\(J\)	\(K\)	\(T\)	\(D\)
0	**0**	0	\(imes\)	**0**	**\(imes\)**	0	0
0	**1**	1	0	**1**	**\(imes\)**	1	1
1	**0**	0	1	**\(imes\)**	**1**	1	0
1	**1**	\(imes\)	0	**\(imes\)**	**0**	0	1

3.1 Universal Flip-Flop Conversion Algorithm (3 Steps)

1. **Identify:** Define the *Available Flip-Flop* (what you have) and the *Required Flip-Flop* (what you want to build).
2. **Construct Combined Truth-Excitation Table:** List columns: [Required Inputs] \(ightarrow\) [Present State \(Q_n\)] \(ightarrow\) [Next State \(Q_{n+1}\)] \(ightarrow\) [Available Inputs needed from Excitation Table].
3. **Minimize using K-Maps:** Solve for each Available input variable in terms of Required inputs and \(Q_n\). Draw the circuit.

Solved Example 1: Convert S-R to J-K

- Available: S-R Flip-Flop; Required: J-K Flip-Flop.
- Excitation extraction gives:
  For \(S\): minterms at \(J=1, □ar{Q}=1\) → **\(S = J□ar{Q}\)**
  For \(R\): minterms at \(K=1, Q=1\) → **\(R = KQ\)**
- **Implementation:** Two 2-input AND gates driving S and R with cross-coupled feedback from \(□ar{Q}\) and \(Q\).

Solved Example 2: Convert J-K to D-Type

- Available: J-K Flip-Flop; Required: D Flip-Flop.
- For \(D=0\): Need \(Q^+=0\) → \(J=0, K=1\)
- For \(D=1\): Need \(Q^+=1\) → \(J=1, K=0\)
- **Equations: \(J = D\)**, and **\(K = □ar{D}\)**
- **Implementation:** Single inverter (7404) between J and K input terminals.

4. Timing Specifications & VLSI Hardware Physics

In real semiconductor circuits, signals take time to propagate, and flip-flops require signals to be stable around clock transitions to prevent **metastability**.

PARAMETER	SYMBOL	DEFINITION	74LS74A (TTL)	74HC74A (CMOS)
Setup Time	t_s	Minimum time data input must remain stable *before* active clock edge	20 ns	14 ns
Hold Time	t_h	Minimum time data input must remain stable *after* active clock edge	5 ns	3 ns
Propagation Delay	t_{pLH}, t_{pHL}	Delay from 50% point of clock transition to 50% point of output Q transition	25 - 40 ns	17 - 18 ns
Clock Pulse Width	t_{CH}, t_{CL}	Minimum duration clock pulse must remain HIGH or LOW	25 ns	10 ns
Max Clock Frequency	f_{max}	Highest rate at which flip-flop can be reliably clocked	25 MHz	35 MHz

> ⏱️ **Clock Skew in High-Speed VLSI Systems**
>
> **Definition:** Difference in clock arrival time at two different flip-flops ($\Delta t_{skew} = t_{clk2} - t_{clk1}$). If destination flip-flop receives clock later than source, a data race occurs. To prevent hold time violations: **$t_{prop(source)} \ge t_{hold(dest)} + t_{skew}$**.

5. Shift Registers & IC 74194

A register is a cascade of flip-flops used for bit storage and bit manipulation. For an \(N\)-bit register, \(N\) flip-flops are required.

5.1 The 4 Modes of Shift Register Operation

MODE	FULL NAME	INPUT TYPE	OUTPUT TYPE	CLOCKS TO WRITE	CLOCKS TO READ	PRIMARY REAL-WORLD USE
SISO	Serial-In Serial-Out	1 bit/clock	1 bit/clock	\(N\) pulses	\(N-1\) pulses	Digital Delay Line: \(\Delta t = N imes rac{1}{f_c}\)
SIPO	Serial-In Parallel-Out	1 bit/clock	All \(N\) bits	\(N\) pulses	0 (Instantaneous)	UART / SPI Receiver (Deserializer)
PISO	Parallel-In Serial-Out	All \(N\) bits	1 bit/clock	1 pulse (load)	\(N\) pulses	UART / SPI Transmitter (Serializer)
PIPO	Parallel-In Parallel-Out	All \(N\) bits	All \(N\) bits	1 pulse	0 (Instantaneous)	CPU Data Bus Buffer / Scratch Register

5.2 IC 74194 (4-Bit Bidirectional Universal Shift Register)

The industry standard MSI universal shift register. Capable of all 4 modes plus bidirectional shifting.

MODE SELECT		CLOCK (\ (CK\))	CLEAR (\ (□AR{CR}\))	OPERATION PERFORMED
\(S_1\)	\(S_0\)			
\(imes\)	\(imes\)	\(imes\)	0	**Reset to 0000 (Asynchronous Clear)**
0	0	\(imes\)	1	Inhibit Clock (Hold State)
0	1	↑	1	**Shift Right (\(D_{SR} ightarrow Q_A ightarrow Q_B ightarrow Q_C ightarrow Q_D\))**
1	0	↑	1	**Shift Left (\(Q_A \leftarrow Q_B \leftarrow Q_C \leftarrow Q_D \leftarrow D_{SL}\))**
1	1	↑	1	Parallel Load (\(A, B, C, D ightarrow Q_A, Q_B, Q_C, Q_D\))

5.3 Ring Counter

- **Connection:** Serial output \(Q_0\) fed back to serial input \(D\) of first stage.
- **Pre-load:** Initialized to \(1000\) (single 1 circulating).
- **Modulus: \(N\) states** for \(N\) flip-flops.
- **Waveform:** Sequential, non-overlapping pulses.
- **Application:** Multiphase clock generator, stepper motor phase sequencer.

5.4 Johnson (Twisted-Ring) Counter

- **Connection:** Inverted serial output \(□ar{Q}_0\) fed back to serial input \(D\).
- **Pre-load:** Starts from cleared state \(0000\).
- **Modulus: \(2N\) states** for \(N\) flip-flops (Twice the states!).
- **Decoding:** Decoded with simple 2-input AND gates without glitch hazards.
- **Application:** High-speed frequency divider, Quadrature clock generation.

6. Counters: Asynchronous (Ripple) & Synchronous Design

6.1 Ripple Counters vs. Synchronous Counters

FEATURE	ASYNCHRONOUS (RIPPLE) COUNTER	SYNCHRONOUS COUNTER
Clock Connection	Only FF0 connected to clock; succeeding FFs clocked by previous output	All flip-flops connected to the exact same master clock simultaneously
Propagation Delay	Cumulative: $t_{total} = N \cdot t_{pd}$	Fixed: Single flip-flop delay t_{pd}
Maximum Frequency	Low: $f_{max} \le rac{1}{N \cdot t_d + T_s}$	High: $f_{max} \le rac{1}{t_d + t_{comb}}$
Decoding Glitches	Severe false spikes due to intermediate transient states	Minimal / Glitch-free

6.2 Key Asynchronous Counter ICs: 7490 & 7493

IC 7490 BCD Decade Counter

- Contains independent Mod-2 (FF_A) and Mod-5 (FF_B, FF_C, FF_D) sections.
- **Mod-10 Connection:** Connect Q_A output to Input B. Apply clock to Input A.
- **Mod-6 Counter Design:** Connect outputs Q_B and Q_C (count $0110 = 6$) to Reset pins R_1, R_2.

IC 7493 4-Bit Binary Counter

- Contains Mod-2 (FF_A) and Mod-8 (FF_B, FF_C, FF_D) sections.
- **Mod-16 Connection:** Connect Q_A output to Input B. Apply clock to Input A. Counts 0 to 15.
- **Mod-12 Counter Design:** Connect outputs Q_C and Q_D (count $1100 = 12$) to Reset pins R_1, R_2.

6.3 Step-by-Step Synchronous Counter Design Procedure

1. Find number of flip-flops required: $2^N \ge M$ (where M is modulus).
2. Write state transition table ($Q_2 Q_1 Q_0 ightarrow Q_2^+ Q_1^+ Q_0^+$).
3. Use Flip-Flop Excitation Table to determine required inputs for each state transition.
4. Plot K-maps for each input and minimize expressions.
5. Draw the circuit diagram.

📐 Master Derived Equations: 3-Bit Synchronous Binary UP Counter (using J-K)

- Stage 0 (LSB): $\mathbf{J_0 = 1, \quad K_0 = 1}$ (Toggles on every single clock pulse)
- Stage 1: $\mathbf{J_1 = Q_0, \quad K_1 = Q_0}$ (Toggles when $Q_0 = 1$)
- Stage 2 (MSB): $\mathbf{J_2 = Q_0 \cdot Q_1, \quad K_2 = Q_0 \cdot Q_1}$ (Toggles when both $Q_0 = 1$ and $Q_1 = 1$)

7. Finite State Machines (FSMs) & Sequential Synthesis

Mealy Model

Outputs are a function of **both present state and current inputs**: \(Y = f(PS, X)\). Outputs change asynchronously if inputs change between clock pulses.

Moore Model

Outputs are a function of **present state only**: \(Y = f(PS)\). Outputs change strictly synchronously on clock edges, eliminating input glitch hazards.

7.1 Solved Classic Design: Sequence Detector for `1010`

Detects the bit stream **`1010`** on a serial input \(X\). When detected, output \(Y = 1\).

STATE	SEQUENCE STORED	NEXT STATE (\(X=0\))	NEXT STATE (\(X=1\))	OUTPUT \(Y\) (\(X=0\))	OUTPUT \(Y\) (\(X=1\))
A (00)	No bit matched (Reset)	A	B	0	0
B (01)	Matched `1`	C	B	0	0
C (10)	Matched `10`	A	D	0	0
D (11)	Matched `101`	C	B	**1 (Sequence 1010 detected!)**	0

📐 Synthesized D Flip-Flop Excitation Equations (from K-Maps)

- Flip-Flop 1 Input: **\(D_1 = Q_0 \bar{X} + Q_1 \bar{Q}_0 X\)**
- Flip-Flop 0 Input: **\(D_0 = X\)**
- Output Equation: **\(Y = Q_1 Q_0 \bar{X}\)**

8. Real-World Practical Applications & Circuit Engineering

8.1 Switch Debouncing Circuit (Chatterless Switch)

Mechanical switches bounce upon closing for 5–20 ms, creating false pulse bursts that cause counters to advance multiple counts. An **S-R latch with pull-up resistors and SPDT push button** completely eliminates bouncing: the first physical contact flip sets the latch, and subsequent chattering produces no state change.

8.2 Stepper Motor 4-Phase Drive Generator

A 4-bit Ring Counter running on clock pulses generates the sequence \(\Phi_1 ightarrow \Phi_2 ightarrow \Phi_3 ightarrow \Phi_4\) (\(1000 ightarrow 0100 ightarrow 0010 ightarrow 0001\)), directly sequencing power transistors to rotate stepper motor poles.

8.3 1-Bit Static RAM (SRAM) Cell Architecture

A clocked D latch combined with address select line (\(A_n\)) and Read/Write control line (\(R/□ar{W}\)): When \(A_n = 1\) and \(R/□ar{W} = 0\), data is written into the cell. When \(A_n = 1\) and \(R/□ar{W} = 1\), data is read out non-destructively.